FOR LOOPS

LEVEL 5: GCD, LCM & Advanced Number Theory

Program 51: GCD of Two Numbers (Euclidean Algorithm) Write a program that finds the GCD of two numbers using the Euclidean algorithm with a loop.

Program 52: LCM of Two Numbers Write a program that finds the LCM of two numbers using the formula: LCM = (a × b) / GCD(a, b).

Program 53: Check if Two Numbers are Co-Prime Write a program that takes two numbers and checks if they are co-prime (GCD = 1).

Program 54: Strong Number Checker Write a program that checks if a number is a Strong number (sum of factorials of digits = number). Example: 145 = 1! + 4! + 5!.

Program 55: Print All Strong Numbers (1 to N) Write a program that finds and prints all Strong numbers from 1 to N.

Program 56: Check if Special Number (Sum of Digits + Product of Digits = Number) Write a program that checks if sum of digits plus product of digits equals the number.

Program 57: Print All Divisor Pairs Write a program that prints all divisor pairs of a number. Example: 12 → (1,12), (2,6), (3,4).

Program 58: Check if Number Can be Expressed as Sum of Two Primes Write a program that checks if an even number (>2) can be expressed as sum of two prime numbers (Goldbach's conjecture verification).

LEVEL 6: Series & Sequences

Program 59: Fibonacci Series (First N Terms) Write a program that prints the first N terms of the Fibonacci series (0, 1, 1, 2, 3, 5, 8, 13, ...).

Program 60: Sum of Fibonacci Series Write a program that calculates the sum of first N Fibonacci numbers.

Program 61: Check if Number is Fibonacci Number Write a program that checks if a given number exists in the Fibonacci series by generating terms until reaching or exceeding the number.

Program 62: Nth Fibonacci Number Write a program that finds and displays the Nth term of Fibonacci series.

Program 63: Arithmetic Progression - Print Series Write a program that takes first term (a), common difference (d), and N, then prints first N terms of AP.

Program 64: Arithmetic Progression - Sum of Series Write a program that calculates the sum of first N terms of an AP using a loop.

Program 65: Geometric Progression - Print Series Write a program that takes first term (a), common ratio (r), and N, then prints first N terms of GP.

Program 66: Geometric Progression - Sum of Series Write a program that calculates the sum of first N terms of a GP using a loop.

Program 67: Sum of Series: 1 + 1/2 + 1/3 + ... + 1/N Write a program that calculates the sum of the harmonic series up to N terms.

Program 68: Sum of Series: 1 - 2 + 3 - 4 + 5 - 6 + ... ± N Write a program that calculates this alternating series.

Program 69: Sum of Series: 1/1! + 1/2! + 1/3! + ... + 1/N! Write a program that calculates this factorial series (approximates e-1).

Program 70: Tribonacci Series (First N Terms) Write a program that prints first N terms of Tribonacci series where each term is sum of previous 3 terms. Start: 0, 0, 1, 1, 2, 4, 7, 13...

Program 71: Sum of Series: $1^2 - 2^2 + 3^2 - 4^2 + ... \pm N^2$ Write a program that calculates alternating sum of squares.

LEVEL 7: Pattern Printing - Numbers

Program 72: Right Triangle - Numbers (Rows)

```
1
1 2
1 2 3
```

1 2 3 4
Write a program that prints this pattern for N rows.

Program 73: Right Triangle - Numbers (Continuous)

1
2 3
4 5 6
7 8 9 10

Write a program that prints continuous numbers in triangle pattern.

Program 74: Right Triangle - Same Number in Row

1
2 2
3 3 3
4 4 4 4

Write a program that prints this pattern for N rows.

Program 75: Right Triangle - Reverse Numbers

1
2 1
3 2 1
4 3 2 1

Write a program that prints this pattern for N rows.

Program 76: Inverted Right Triangle - Numbers

1 2 3 4
1 2 3
1 2
1

Write a program that prints this inverted pattern for N rows.

Program 77: Number Pyramid (Centered)

```
   1
  1 2
 1 2 3
1 2 3 4
```

Write a program that prints this centered number pyramid for N rows.

Program 78: Number Diamond

```
   1
  1 2
```

```
  1 2 3
   1 2
    1
```
Write a program that prints a diamond pattern with numbers for N rows.

Program 79: Inverted Number Pyramid

```
1 2 3 4
 1 2 3
  1 2
   1
```
Write a program that prints inverted number pyramid for N rows.

Program 80: Hollow Square with Numbers

```
1 2 3 4
1     4
1     4
1 2 3 4
```
Write a program that prints a hollow square with numbers on borders for N×N size.

Program 81: Palindrome Number Pattern

```
1
121
12321
1234321
```
Write a program that prints palindrome number pattern for N rows.

Program 82: Reverse Palindrome Pattern

```
1234321
 12321
  121
   1
```
Write a program that prints inverted palindrome number pattern for N rows.

Program 83: Binary Number Pattern

```
1
0 1
1 0 1
0 1 0 1
```

Write a program that prints alternating 0 and 1 in triangle pattern for N rows.

Program 84: Multiplication Table Grid (NxN) Write a program that prints a multiplication table grid from 1 to N (N×N table).

Program 85: Hourglass Number Pattern

```
1 2 3 4 5
 2 3 4 5
  3 4 5
   4 5
    5
   4 5
  3 4 5
 2 3 4 5
1 2 3 4 5
```

Write a program that prints hourglass pattern with numbers for N rows.

Program 86: Square Border Pattern with Numbers

```
1 1 1 1 1
1 2 2 2 1
1 2 3 2 1
1 2 2 2 1
1 1 1 1 1
```

Write a program that prints this square border pattern for N×N.

Program 87: Number Zigzag Pattern

```
1
 2 3
   4 5 6
     7 8 9 10
```

Write a program that prints zigzag number pattern for N rows.

Program 88: Floyd's Triangle

```
1
2 3
4 5 6
7 8 9 10
```

Write a program that prints Floyd's triangle for N rows.

LEVEL 8: Pattern Printing – Stars

Program 89: Right Triangle - Stars

```
*
* *
* * *
* * * *
```

Write a program that prints this star pattern for N rows.

Program 90: Inverted Right Triangle - Stars

```
* * * *
* * *
* *
*
```

Write a program that prints this inverted star pattern for N rows.

Program 91: Pyramid - Stars (Centered)

```
   *
  * *
 * * *
* * * *
```

Write a program that prints centered pyramid with stars for N rows.

Program 92: Inverted Pyramid - Stars

```
* * * *
 * * *
  * *
   *
```

Write a program that prints inverted centered pyramid for N rows.

Program 93: Diamond - Stars

```
   *
  * *
 * * *
  * *
   *
```

Write a program that prints a diamond pattern with stars for N rows.

Program 94: Hollow Square - Stars

```
* * * *
```

```
*      *
*      *
* * * *
```

Write a program that prints hollow square with stars for N×N size.

Program 95: Hollow Pyramid

```
   *
  * *
 *   *
* * * *
```

Write a program that prints hollow pyramid (only borders have stars) for N rows.

Program 96: Hollow Diamond

```
   *
  * *
 *   *
*     *
 *   *
  * *
   *
```

Write a program that prints hollow diamond pattern for N rows.

Program 97: Hollow Inverted Pyramid

```
* * * * *
 *     *
  *   *
   * *
    *
```

Write a program that prints hollow inverted pyramid for N rows.

Program 98: Butterfly Pattern

```
*         *
* *     * *
* * * * * *
* *      * *
*          *
```

Write a program that prints butterfly pattern with stars for N rows (must be odd).

Program 99: Sandglass Pattern

```
* * * * *
 * * * *
  * * *
   * *
    *
   * *
  * * *
 * * * *
* * * * *
```

Write a program that prints sandglass pattern for N rows.

Program 100: Rhombus Pattern

```
    * * * *
   * * * *
  * * * *
 * * * *
```

Write a program that prints rhombus pattern for N rows.

Program 101: Hollow Rhombus

```
    * * * *
   *     *
  *     *
 * * * *
```

Write a program that prints hollow rhombus pattern for N rows.

Program 102: Cross (X) Pattern

```
*       *
 *     *
  *   *
   * *
    *
   * *
  *   *
 *     *
*       *
```

Write a program that prints X pattern for N rows (must be odd).

Program 103: Arrow Pattern (Upward)

```
    *
```

```
   * *
  * * *
 * * * *
    *
    *
    *
```

Write a program that prints upward arrow pattern for N rows.

Program 104: Hollow Rectangle - Stars

```
* * * * *
*       *
*       *
* * * * *
```

Write a program that prints hollow rectangle with M rows and N columns.

Program 105: Square Within Square

```
* * * * *
* + + + *
* + * + *
* + + + *
* * * * *
```

Write a program that prints square within square pattern for N×N (use different characters).

Program 106: Pascal Triangle Shape (Stars Only)

```
    *
   * *
  * * *
 * * * *
* * * * *
```

Write a program that prints Pascal triangle shape with stars for N rows.

Program 107: Left Triangle - Stars

```
      *
    * *
  * * *
* * * *
```

Write a program that prints left-aligned triangle for N rows.

Program 108: Inverted Left Triangle - Stars

```
* * * *
  * * *
    * *
      *
```

Write a program that prints inverted left-aligned triangle for N rows.

Program 109: Plus (+) Sign Pattern

```
    *
    *
* * * * *
    *
    *
```

Write a program that prints plus sign pattern with stars for size N.